

Customer Churn Prediction

Project Description

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1. Project Overview

Customer churn — the loss of clients or customers — is one of the most critical challenges facing telecom companies. Once a customer decides to leave, recovering them is significantly more expensive than retaining them proactively. This project addresses that challenge by building a complete, end-to-end Machine Learning system capable of predicting which customers are likely to churn before they do.

By analyzing customer demographics, financial data, service usage, complaint history, and satisfaction scores, our system identifies high-risk customers and provides actionable insights to business teams — enabling targeted retention campaigns that save revenue and improve customer loyalty.

2. Problem Statement

A major telecom company is losing approximately 9.9% of its customer base annually to churn. With a dataset of 999,999 customers, the company needs to:

- Identify which customers are at risk of leaving before they actually leave
- Understand the key factors driving churn (complaints, contract type, satisfaction)
- Enable business teams to take proactive retention actions with data-driven prioritization
- Provide a real-time prediction tool accessible to non-technical users

3. Proposed Solution

We built a full ML pipeline consisting of five integrated components:

Component	Description	Owner
Data Engineering	Clean, encode, and scale 999,999 customer records into model-ready format	Data Engineer
Exploratory Analysis	Identify churn patterns across contract type, tenure, complaints, satisfaction	Data Analyst
SQL Database	Load data into SQL Server; run 5 business intelligence queries	SQL Engineer
ML Modeling	Train & evaluate 4 models; select LightGBM as best performer	Data Scientist
Streamlit Dashboard	Live web app with overview, EDA charts, and real-time prediction tool	BI Developer

4. Dataset

The dataset used is a synthetic telecom customer churn dataset with the following characteristics:

Property	Value
Total Records	999,999 customers
Total Features	32 columns
Target Variable	churn (0 = Stayed, 1 = Left)
Churn Rate	9.92% — significant class imbalance (9:1 ratio)
Missing Values	5 columns with < 5% missing data
Feature Categories	Demographics, Financial, Services, Behavioral, Risk Indicators

Key features include: tenure, contract type, monthly charges, number of complaints, customer satisfaction score, credit score, days since last interaction, and 8 binary service subscription columns.

5. Methodology

5.1 Data Preprocessing

- Dropped non-predictive columns: customer_id, signup_date
- Filled 5 columns with missing values using Median imputation
- Applied Ordinal Encoding to: contract, education, paperless_billing
- Applied One-Hot Encoding to: gender, marital_status, payment_method
- Applied StandardScaler to 13 numerical columns
- Split data: 80% training (799,999 rows) / 20% testing (200,000 rows) with stratification

5.2 Model Training & Selection

Four classification models were trained and evaluated using class_weight='balanced' to handle the 9:1 class imbalance:

Model	F1-Score (Churn)	Recall	ROC-AUC	Result
LightGBM	0.257	63.6%	0.635	✔ Selected
Logistic Regression	0.256	64.2%	0.635	Runner-up
XGBoost	0.253	60.3%	0.627	Tested
Random Forest	0.001	0.0%	0.500	Needs tuning

LightGBM was selected for its best F1-Score, strong ROC-AUC, and superior speed on 1M+ row datasets. F1-Score was prioritized over Accuracy due to the significant class imbalance.

5.3 SQL Analysis

The dataset was loaded into Microsoft SQL Server (OMAR-ZIZO, version 16.0). Five analytical queries were executed:

- Overall churn rate: 9.92% across 999,999 customers
- Churn by contract: Month-to-Month 26.51% vs Two-Year 5.65%
- High-risk customer identification: 100 retained customers with warnings
- Churn by monthly charges: Minimal variation across billing tiers
- Churn by tenure: New customers (<1yr) at 10.26% vs loyal (3yr+) at 9.33%

6. Key Insights

Insight	Finding	Business Action
Contract Type	Month-to-Month churn at 26.5% vs 5.6% for Two-Year	Offer discounts to switch to annual plans
Customer Tenure	New customers (0-10 months) churn most	Launch onboarding & early loyalty programs
Complaints	Churned customers have significantly more complaints	Resolve complaints within 24 hours
Satisfaction	Low satisfaction scores predict churn reliably	Regular surveys + proactive follow-up
Monthly Charges	Very High charges = 12% churn rate (highest tier)	Review pricing for premium segments

7. Live Application

A Streamlit web application was built to make the ML model accessible to business users without technical knowledge. The app runs at localhost:8501 and consists of three pages:

Page	Features
 Overview	4 KPI metric cards + Key Insights summary panel
 EDA	4 dark-themed interactive charts: churn distribution, contract type, complaints, satisfaction
 Prediction	9 input sliders + real-time LightGBM prediction + risk probability + auto-detected risk factors

8. Technology Stack

Category	Tools & Libraries
Language	Python 3.14.1
Data Processing	Pandas 2.2.3, NumPy 2.1.3
Machine Learning	Scikit-learn 1.5.2, LightGBM 4.5.0, XGBoost 2.1.3
Class Imbalance	Imbalanced-learn 0.12.4 (SMOTE)
Visualization	Matplotlib 3.9.3, Seaborn 0.13.2
Database	Microsoft SQL Server 16.0, pyodbc 5.2.0
Deployment	Streamlit 1.40.2
Model Saving	joblib 1.4.2

Category	Tools & Libraries
IDE & Tools	VS Code, Jupyter Notebooks, Git

9. Project Team

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